# Differential enrichment of yeast DNA in SARS-CoV-2 and related genomes supports synthetic origin hypothesis


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## Introduction and main hypothesis

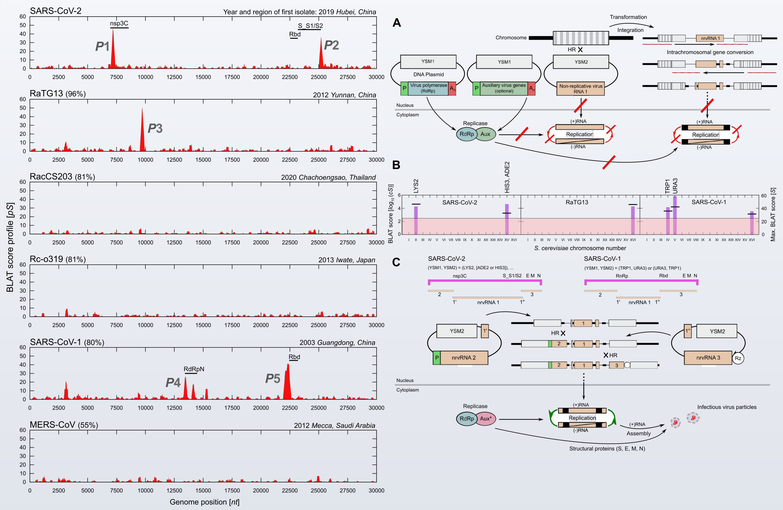
Already in March 2020 evidence was put forward that the pandemic outbreak of novel coronavirus SARS-CoV-2 within the human population was most likely a product of natural evolution. According to this view, COVID-19 is a zoonosis that originated from a species of closely related bat coronaviruses. Prior to a hypothetical spillover event, a recent ancestor to SARS-CoV-2 likely evolved inside bat host cells for many decades.

However, natural evolution as an exclusive explanation of SARS-CoV-2 origin and biogenesis is currently not without considerable limitations: the difficulty in characterizing the evolutionary origin of the unusual poly-basic (RRAR) furin cleavage site at the S1/S2 junction of the SARS-COV-2 spike (S) glycoprotein; the discrepancy between a strongly suppressed tropism of SARS-CoV-2 in *Rhinolophus sinicus* cells and the relatively high susceptibility of SARS-CoV-2 toward cell entry via *Rhinolophus sinicus* angiotensin-converting enzyme 2, its primary entry receptor; and the persistent inability to identify an intermediate ancestral host between human and the horseshoe bat *Rhinolophus affinis*. This species was reported to be the host of coronavirus RaTG13, currently the isolate with the highest sequence similarity to the SARS-CoV-2 genome, which is located on the same clade as *Rhinolophus sinicus* bat coronavirus Finding the last animal progenitor host of SARS-CoV-2 has been further complicated by a continued uncertainty about the origin of RaTG13 itself.

In contrast to the natural evolution hypothesis, the above limitations do not necessarily apply to genetic engineering of viral genomes. For example, in the case of SARS coronavirus, it has long been established that introducing a synthetic poly-arginine construct at the furin cleavage site significantly increases the rate of entry into human cells compared with wild-type spike protein. Also before 2010, after a period of rapid progress in the understanding the relevant host-virus factors, natural barriers in host range of positive strand RNA viruses were rationally extended, leading to directed viral replication in new species including model organisms that originally were not permissive, such as the yeast *Saccharomyces cerevisiae.* (Alves-Rodrigues et al. 2006). Accordingly, to transform yeast into a synthetic host, the scheme has been to co-express viral RNA dependent RNA polymerase (RdRp) and, optionally, additional viral factors on plasmids under the control of auxotrophic yeast selectable markers (YSMs). YSMs are primarily there to direct cell lines into stable expression of desired plasmid DNA, but at the same time may function as entry gates for exogenous genetic material into yeast chromosomes (Compton *et al*. 1982).

In principle, this approach applies to any replication competent SARS coronavirus RNA, including cloned or *de novo* synthesized genomic parts or even entire genomes, thus facilitating their replication as well as integration into the yeast genome. ***Our hypothesis is that such a passage would leave behind traces in the genomes of both the virus construct and the synthetic host.***

## Results – Homology signals and artificial passage model



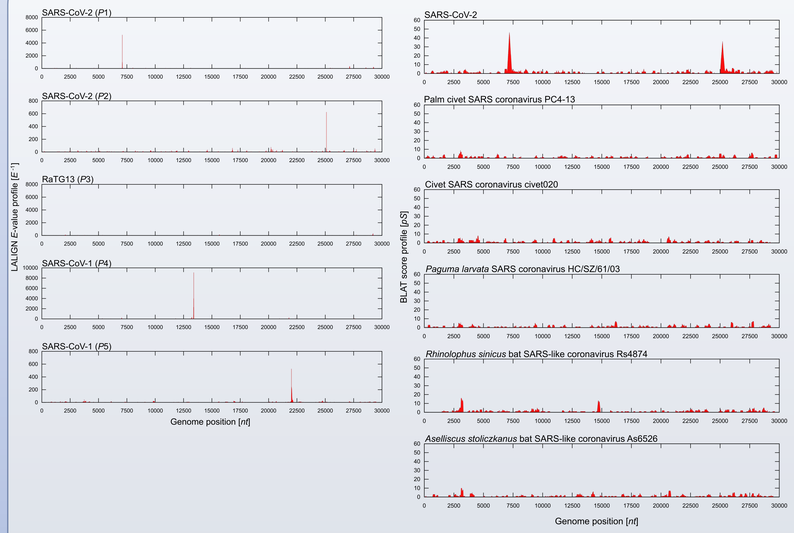

(***Left***) Profiled BLAT scores (*pS*) from the alignment output to the query input of six SARS-coronavirus related full genome sequences show five prominent ($pS > 20$) peaks (*P*1-*P*5). Alignment scores retrieved from BLAT best hits matching *S. cerevisiae* genomic sequences.

(***Right* A**) First stage of passage model in the artificial host *S. cerevisiae* of a plasmid encoded, non-replicable viral RNA (nrvRNA 1) originating from a SARS-CoV related virus. Primary integration of non-homologous nrvRNA sequence occurs through homologous recombination (HR) between the auxotrophic plasmid yeast selectable marker YSM1 (grey box) and its chromosomal homolog (striped grey box). This is followed by higher-order homologous recombination on the flanking regions of nrvRNA1 through intra-chromosomal gene-conversion. Co-expression of viral replicase complex through plasmid expression of RdRp and auxiliary viral genes (*e.g.*, nsp3C, nsp4). Graphical scheme in parts adapted from Compton *et al*. (1982), and from Alves-Rodrigues *et al*. (2006). P, yeast promoter; An, poly-adenosine sequence.

(***Right B***) Integrated profile scores, *cS*, from BLAT sequence hits on S. cerevisiae by chromosome number from the same six SARS-CoV related genomic input sequences (purple columns); a score profile cutoff $pS > 30$ was used. Without a cut-off, the same order emerges (black horizontal bars; all other maximum *pS* scores from the other genomic queries are below, within shaded area). Five common yeast selectable markers are assigned to their chromosomes of origin.

(***Right C***) Procedural second stage scheme for the synthetic biogenesis of SARS-CoV-2 and SARS-CoV-1. Given pairings of yeast selectable markers (YSM1, YSM2) matched in (***B***), genome editing of the three segments nrvRNA1, 2, 3 through higher order transformation and HR leads to a fully replicable virus (+)sense RNA. For virus (self-)assembly, the structural outer proteins S, E, M, and N are expressed from an enhanced plasmid set Aux*. Rz, self-cleaving ribozyme.

## Results – Cross-validation and negative control




(***Left***) Cross-validation: Peaks *P*1, *P*2, *P*4, *P*5 (but not *P*3) in alignment *E*-values (inverted, 1/*E*) as profiles across genomes of SARS-CoV-2, RaTG13, and SARS-CoV-1 calculated with the LALIGN local alignment method by using a sliding window approach (window sizes from SARS-CoV alignment region lengths).

(***Right***) Negative control: Smoothed profile BLAT alignment scores of five betacoronavirus isolates from five wild animals, in comparison to SARS-CoV-2 (sequence identity between 79% and 80% to SARS-CoV-2); after the phylogenetic analysis of Li *et al*. (2020).

## Summary


The results reveal a highly differential homology signal on SARS-CoV-2 and SARS-CoV-1 genomes which, according to our model, points to their history of targeted integration, recombination, and directed viral replication through passage in an artificial *S. cerevisiae* host. This genomic pattern suggests a similar synthetic history of SARS-CoV-1 and SARS-CoV-2, but at the same time excludes RaTG13, RacCS203, Rc-o319, and MERS-CoV from this type of genetic origin.


## References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